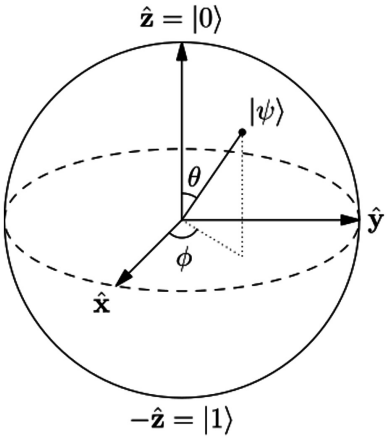
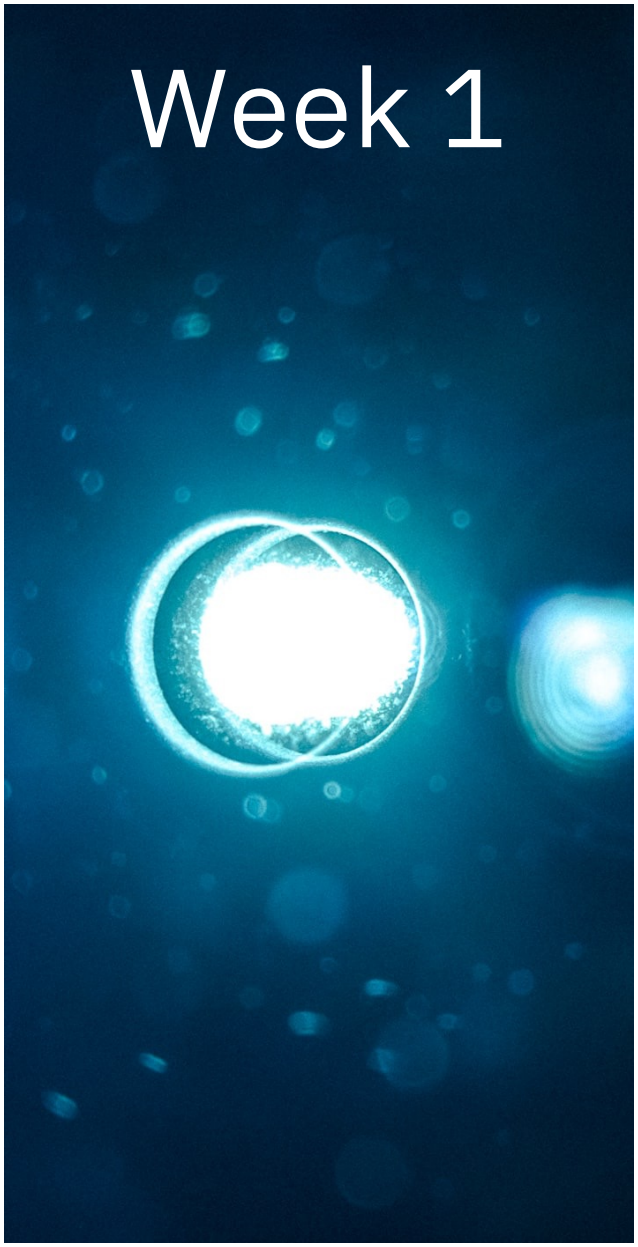
Quantum Coding with Qiskit – A Course for Beginners

13th May 2026

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics





Welcome & Agenda / Week 1

Week 1 — Course Foundations & Environment Setup + finding your Study Group 🚀 / This is a **content week**

- Course Orientation
  - Your Hosts
  - Course philosophy
  - Course structure and expectation

- Introduction to Quantum Computing
  - What is Quantum Computing - IBM Quantum 101
  - What are some of the top use cases for QC
  - NISQ vs Fault-Tolerant Quantum Computing (current limitations, Quantum advantage)

- Introduction to Qiskit
  - What is Qiskit
  - Qiskit ecosystem

- Building Study Groups

- Q&A

- Environment Setup

Course Orientation | Your Hosts



Joey Rogers

Quantum Computing Professional
Algorithm Engineer
Lecturer



Sebastian Grodzietzki

Quantum Computing & Intelligent Automation
Quantum Ambassador Leader DACH, IBM
Developer Ecosystems Leader DACH, IBM
Community Builder & Leader
Tech Go-to-Market
Serial Founder

Course Orientation | Course Philosophy

This is a 3-months course (**13-weeks) with content weeks & study weeks**, focused on **hands-on learning with Qiskit using IBM Quantum hardware.**

Format: Online/Hybrid with selected sessions in the IBM office in Berlin.
Flow: Weekly lectures in first month. Followed by biweekly lectures (new content) and alternating/in-between study group time.
Tools: Python/Qiskit, Jupyter Notebooks, VS Code, IBM Bob, IBM Quantum Platform (but feel free to use whatever you like)
Level: Beginner



This course focuses on:
- Understanding quantum computing intuitively first
- Learning by coding small examples
- Working in study groups
- Building toward real-world applications / Simulating Nature

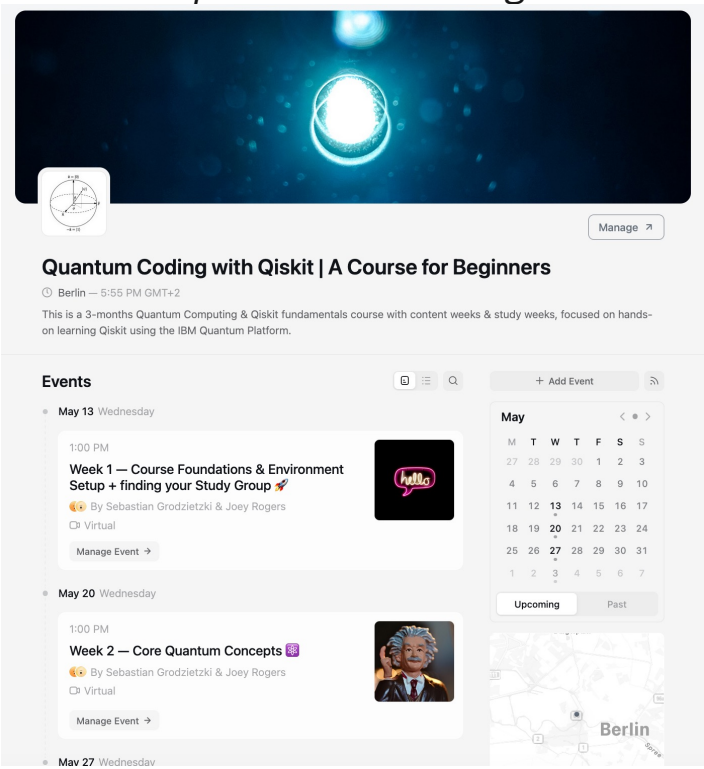
Students will learn:
- Quantum computing concepts
- Quantum circuits
- Qiskit programming
- Variational algorithms adressing quantum chemistry

Learning Outcomes - By the end of the course students will:
- Understand core quantum computing principles
- Build quantum circuits in Qiskit
- Run circuits on simulators and real hardware
- Understand NISQ-era algorithms
- Implement VQE-style workflows (for nature simulations)
- Understand error mitigation basics
- Complete a small real-world quantum project on molecule design

Course Orientation | Course Structure

Each **content week** introduces new material.
Each **study week** focuses on practice and discussion within your group, without Joey & me.

What to expect in the coming weeks…



Study Groups:
Before the first session, we identified who has knowledge in quantum mechanics, physics, chemistry, and coding skills, etc.. We carefully mixed you into teams so each team has a balanced set of experience. ☺

Please, by Friday 10:00, complete the survey:
https://forms.gle/Cy8PieWBQrjXqx8CA
Incl. your email address you want us to share with your small study group so you can reach each other.

Introduction to Quantum Computing

1. What is Quantum Computing - IBM Quantum 101
2. What are some of the top use cases for QC
3. NISQ vs. Fault-Tolerant Quantum Computing (current limitations, Quantum advantage)

IBM Quantum Data Center in Europe – Grand Opening Event

"The opening of the IBM Quantum Data Center in Ehningen is good news for Germany. It will serve as a location for innovation and business growth and is an expression of investors' confidence in the German market. IBM enriches the German quantum computing landscape with this new data center. The German government is providing targeted support for the development of quantum technologies. It is thereby driving forward the development of competencies and capacities in quantum computing in order to promote a robust ecosystem around the development of quantum computers," said **Olaf Scholz, Chancellor of the Federal Republic of Germany.**







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BasQ
Basque Quantum

A regional quantum initiative promoted by the Basque Government for developing industrially useful algorithms

Joint sponsored research in applications in condensed matter, physics and chemistry

Now home to Europe's first IBM Quantum System Two, located at the IBM-Euskadi Quantum Computational Center in San Sebastián



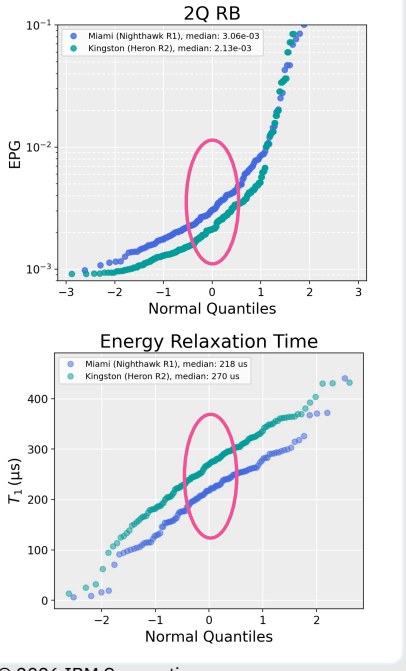




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Our latest processor

IBM Quantum Nighthawk





→ 30% more complex circuits

→ 120 qubits

→ 218 couplers, >20% increase over Heron

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IBM Quantum Mission

→ Bring useful quantum computing to the world

→ Make the world quantum safe



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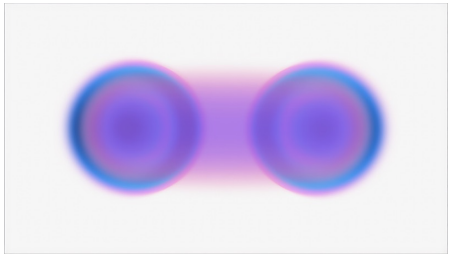
What is quantum computing?

Uniquely quantum

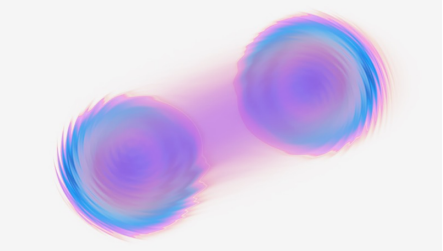
Some problems are classically intractable and will never be solvable with traditional computers



Superposition
A quantum system existing in a complex linear combination of two states until it is measured



Entanglement
Information shared jointly between entangled pairs or groups



Interference
Interaction that affects likelihood of solutions

Moore's law: the number of transistors in a classical integrated circuit doubles about every two years
... but we are approaching the end due to physical limitations

Approaching the physical limit: IBM created the world's first 2 nm node chip in 2021, with transistors as small as 10 silicon atoms

These Quantum concepts can reduce the number of computational steps required for certain algorithms
&
At a ~100 qubit scale, with sufficient circuit depth and complexity, classical computers can no longer simulate exactly

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The new wave of computing



Classical computer
Well suited for many problems



Quantum computer
Unlock classically intractable problems

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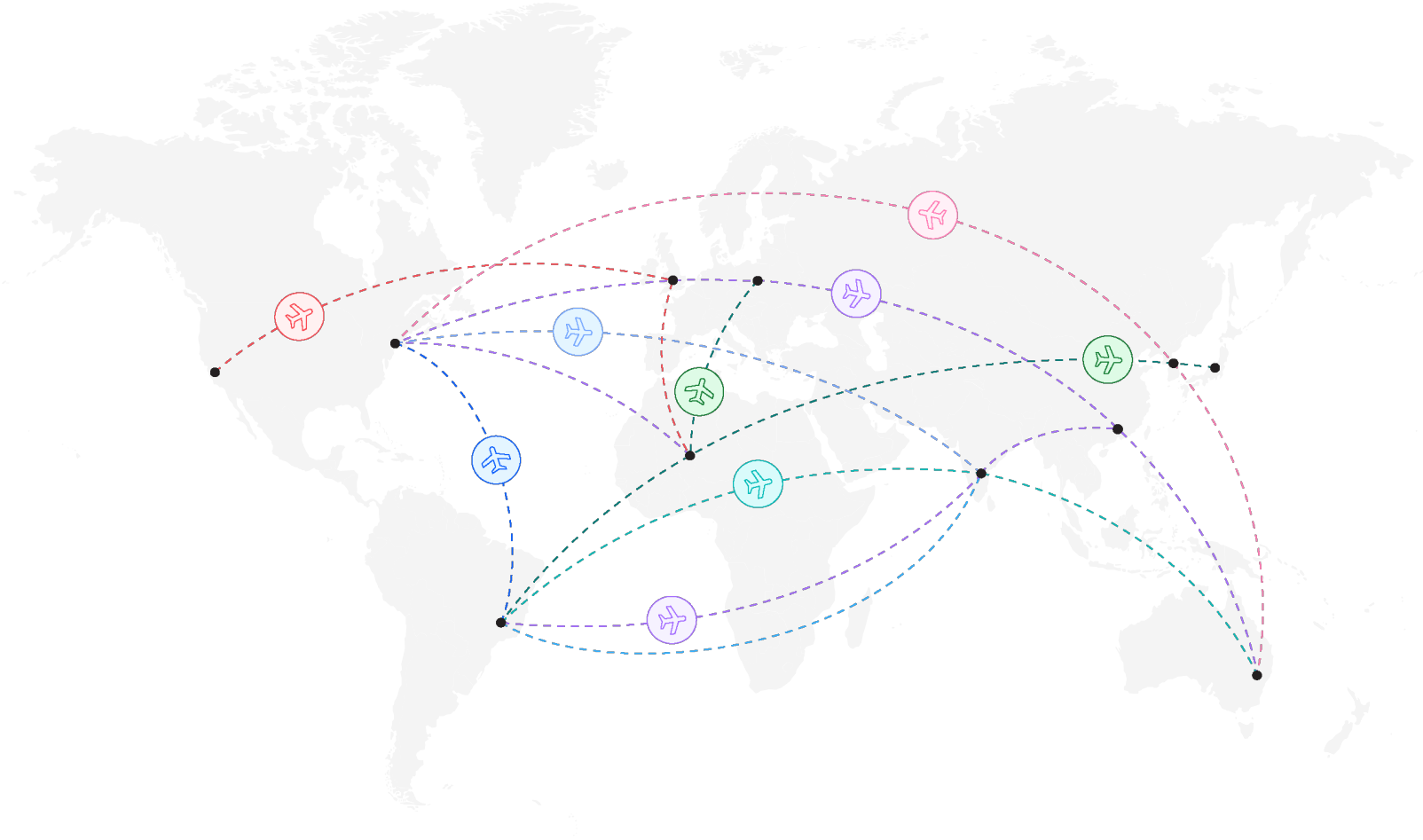
Quantum computing

The new wave of computing

Potential to Unlock previously unsolvable problems with quantum computing, cutting computing time **from years to hours.**

A new paradigm of thinking launches inventors into previously unnavigated discovery territories with **new use cases.**

Accelerate discovery through a **powerful** hybrid quantum-classical approach.



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The practical path to useful quantum computing



Run quantum circuits accurately on quantum hardware

100+ qubit systems and software

Methods that provide execution of quantum circuits on hardware with higher accuracy



Map interesting problems to quantum circuits

Computational tasks that use those circuits

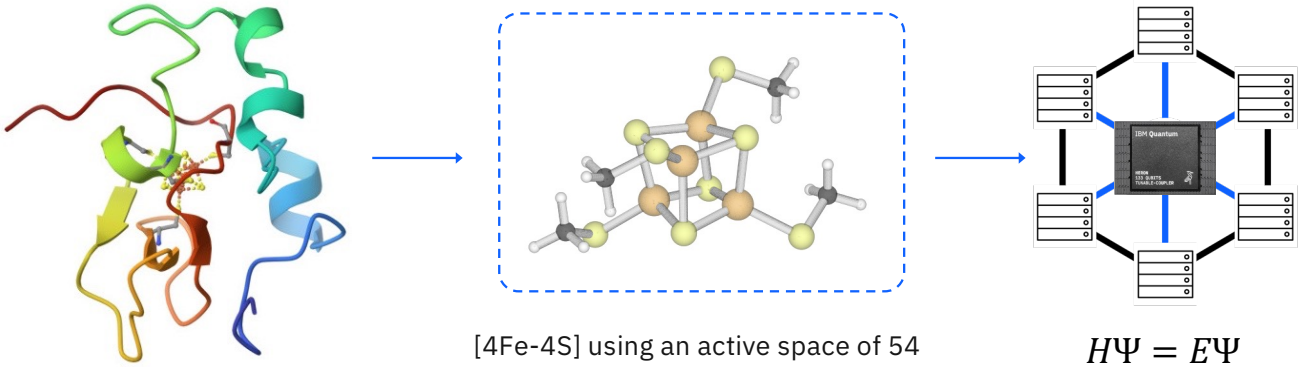
Industry and science applications that use those tasks

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What are these problems?

Simulating Nature

Modeling molecules, atoms, electrons, and quarks with unprecedented accuracy



[4Fe-4S] using an active space of 54 electrons in 36 orbitals from the TZP-DKH basis set

$$H\Psi = E\Psi$$

Classical approximate method (DMRG)	**8 hours** [2,3]
Quantum method (QCSC)	**12 min** using 77 qubits + 1.5 hours supercomputer time[1]

We are already using quantum (Heron, 77 qubits) and HPC (Supercomputer Fugaku) to achieve results comparable with the best classical approximate methods (DMRG) in accuracy and timing

1. Science Advances vol. 11, no. 25 (2025)
2. J. Chem. Theory Comp. 20 (2024): 775–786
3. J. Chem. Phys. 159, 234801 (2023)

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What are these problems?

Modeling molecules, atoms, electrons, and quarks with unprecedented accuracy



Developing lighter, longer-lasting **batteries** for electric vehicles, electronics, and energy grid storage



Designing lighter, stronger **materials** to allow planes to be more efficient and to need less maintenance



Discovering new classes of **antibiotics** to counter the emergence of multidrug-resistant bacterial strains



Designing optimal **superconductors** for MRI, electromobility, and renewable energies

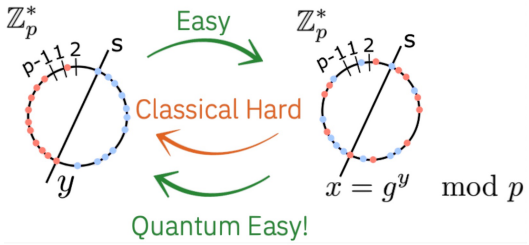
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What are these problems?

Unlock Hidden Patterns

Groups encode symmetries and transformations of data.

Solving algebra in [exponential] spaces.
Finding hidden patterns in structured problems.



$$f(g\mathbf{b}) = f(g) \qquad \forall\ \mathbf{b} \in B$$

$$= \frac{1}{\sqrt{|G|}} \sum_{g \in G} |f(g)\rangle |g\rangle$$

$$= \sqrt{\frac{|B|}{|G|}} \sum_{g_0 \in G/B} |f(g_0)\rangle \frac{1}{\sqrt{|B|}} \sum_{\mathbf{b} \in B} |g_0 \mathbf{b}\rangle$$

Quantum computers **speak the mathematics** of **group theory** and work in the large spaces needed to answer difficult computational problems inaccessible to classical methods.

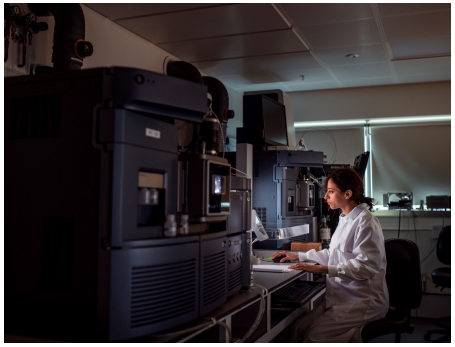
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What are these problems?

Solving algebra in [exponential] spaces.
Finding hidden patterns in structured problems.



Improving understanding of *physical* and *chemical properties* of materials



Improving patient outcomes by designing optimal *cell-centric therapeutics*



Strengthening risk management through better time series and sequence *prediction*



Optimizing vehicle routing and scheduling for *large-scale* logistics networks

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And we know there are other problems with a structure that quantum can exploit.

↳ So we’re also looking for value in problems that are hard classically, like optimization.

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Exploring business value with chemistry and material science

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Quantum applications

Chemical and materials discovery

Read the white paper on quantum-centric supercomputing for materials science
↳ https://arxiv.org/pdf/2312.09733



Chemical and materials science problems are often computationally intensive with exponential resource scaling or rely on approximations that fail for highly correlated systems or become intractable for systems of large enough size.

With primarily quantum data, chemistry and materials science applications may be some of the first to show quantum advantages, particularly when explored through a quantum-centric supercomputing architecture.

We are exploring quantum computing applications in:

- Battery design (e.g. solve the problem where sulfur vanishes)
- Semiconductors, superconductors, and sensors (e.g. to conduct electricity without energy loss)
- Advanced catalysts
- Hydrocarbons and new fuels
- Optimized material structure (e.g., ply composites, corrosion prevention)

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Boeing

Exploring surface interactions in harsh environments

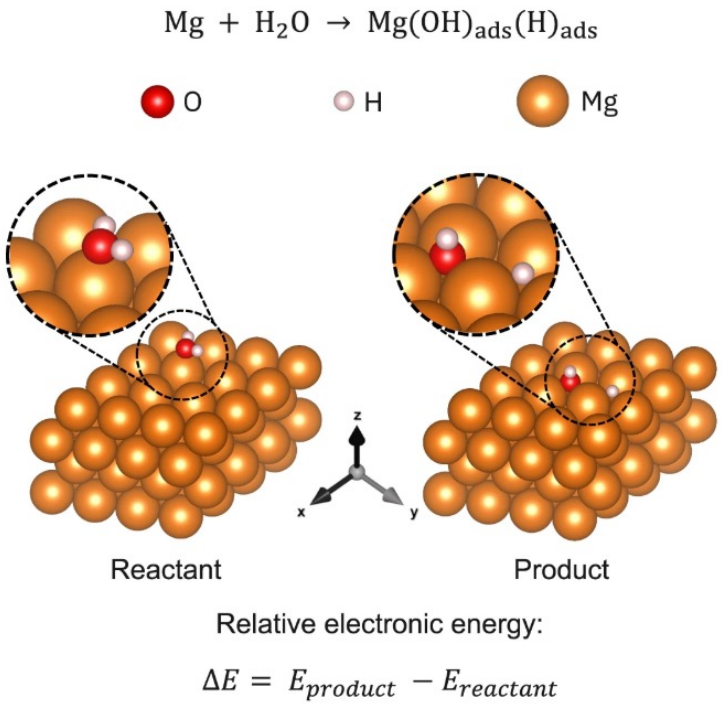
Hamiltonian simulation

Boeing partnered with IBM to develop solutions for material surface reactions simulations relevant in the aerospace industry, focusing first on corrosion in magnesium, with the aim of improving material design and eventually product performance.

Running experiments of up to 20 qubits, the team demonstrated a novel methodology for simulating active spaces in surface material reactions, with application to water reduction in corrosion. The embedding method developed could be used to study other complex periodic systems.

"It shows us that it's not *if* quantum computers will be relevant to our business problems, but *when*."

Jay Lowell
Chief Scientist, Disruptive Computing and Networks, Boeing



- Reaction for splitting of water on a magnesium surface, including schematics of the optimized structures for the reactant and product.

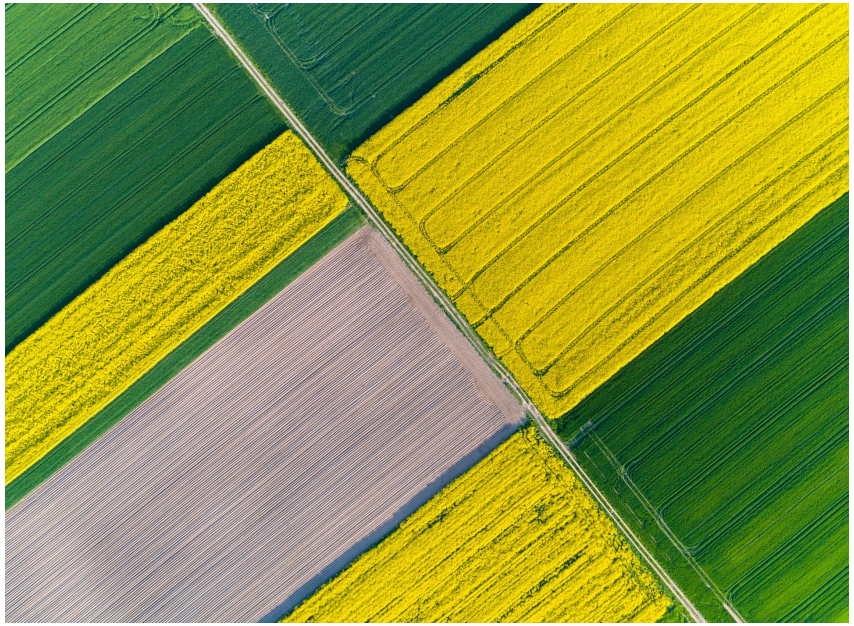
Read the paper: *npj Quantum Information* 9, 88 (2023).

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23

Producing lower-cost, more energy-efficient fertilizer

Hamiltonian simulation



Source 1: “Nitrogen fertiliser production outstrips global needs and exceeds planetary boundaries by factor of two,” Planet Tracker, 14 Nov 2023.

The production of nitrogen fertilizer currently takes ~2% of the world’s energy and is responsible for between 1.4% and 5% of global greenhouse gas emissions.[1]

Using sample-based quantum diagonalization (SQD) methods, IBM and RIKEN have simulated ground states of iron sulfur clusters [2Fe-2S] and [4Fe-4S] that exceed sizes amenable to exact diagonalizations, using up to 77 qubits of the IBM Quantum Heron processor.

Read the paper
↳ arXiv:2405.05068

The biosynthesis of FeMoco (Fe_7MoS_9C), a catalyst for nitrogen fixation, is a very complex process intractable to classical computers. Quantum computers could model this process at a molecular level to design catalysts for the Haber-Bosch process that work at lower temperatures and lower pressures.

As Fe-S is the active center for the nitrogenase enzyme and iron-based catalysts for the Haber-Bosch process, SQD methods could help simulate FeMoco with increased precision, potentially yielding insights that could enhance food security, counter runoff, and increase crop revenue.

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Producing lower-cost, more energy-efficient fertilizer

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↳ arXiv:2405.04068

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Source 1: "Nitrogen fertiliser production outstrips global needs and exceeds planetary boundaries by factor of two," Planet Tracker, 14 Nov 2023.

Fertilizer from the Air / Nitrogen Fixation / The Fe-Mo Cofactor (The Haber-Bosch Dilemma)

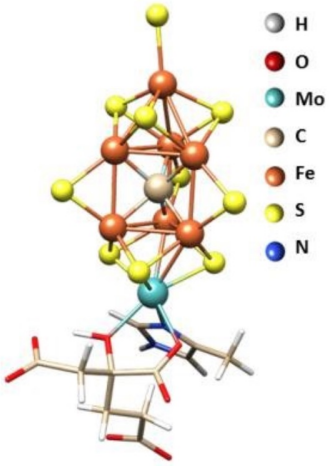


The Problem: Plants need nitrogen to grow. The nitrogen molecule in the air is like an extremely tightly knotted rope which cannot be consumed by plants out of the air. To make fertilizer, we compress air under immense pressure and heat (500 °C and 200 Bar pressure) to untie the knot (split the molecule). This **wastes vast amounts of gas and approximately 2% of the entire world's energy**. Bacteria do this effortlessly at **room temperature** using the enzyme nitrogenase to help the nature grow.

The Solution: We need to find out how by **gaining a precise chemical understanding** of the active Fe-Mo cofactor **to artificially develop** efficient biomimetic **catalysts**.

How: We "recreate" the enzyme in the quantum computer and watch how it unties the knot. We simulate the electronic structure of the FeMo center and the bacteria's "tool" (nitrogenase).

1. We map the molecule to the quantum computer (electrons mapped to qubits)
2. We program the interactions between the iron qubits exactly as they occur in nature.
3. Add nitrogen: We virtually add a nitrogen molecule (N_2) to the cluster (more qubits).
4. Observe the binding: We watch as the electron clouds of the iron shift onto the nitrogen over time and vice versa which releases the strong binding (they exchange electrons)
5. Measure the energy: **We look for the moment when the extremely strong bond of the nitrogen becomes unstable** (and reacts to ammonia which is the most important basic chemical for fertilizer).



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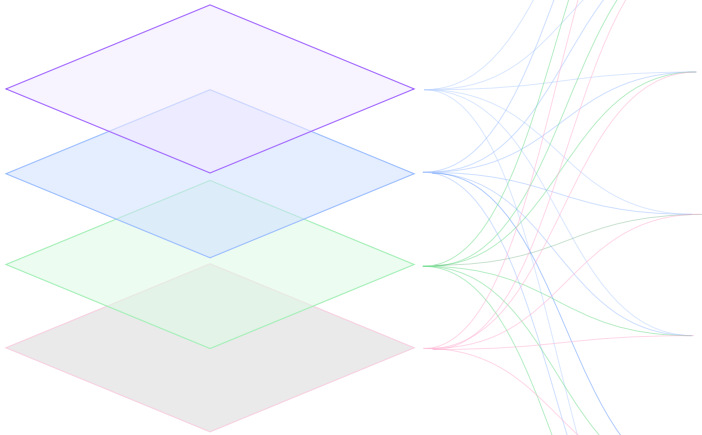
Quantum computing is expected to have impact across industries

Simulating Nature/ Chemistry & materials

Optimization

Quantum machine learning

Partial differential equations (PDEs)



Aerospace	• High strength-to-weight materials development • Wing ply-composite design optimization • Aircraft computational fluid dynamics
Automotive	• Advanced batteries • Multi-objective vehicle routing optimization • Auto design and simulation
Chemicals	• Lower energy fertilizer manufacturing • Efficient catalysts • Accelerated product development
Electronics & semiconductors	• Placement and routing optimization • Deposition/etch process modeling • Multi-domain simulation
Energy	• Grid management • Grid optimization with EVs • Carbon sequestration
Financial services	• Portfolio optimization • Derivatives/options pricing • Anti-money laundering
Healthcare & life sciences	• Drug discovery and drug design optimization • mRNA & protein structure predictions • Disease target & biomarker discovery

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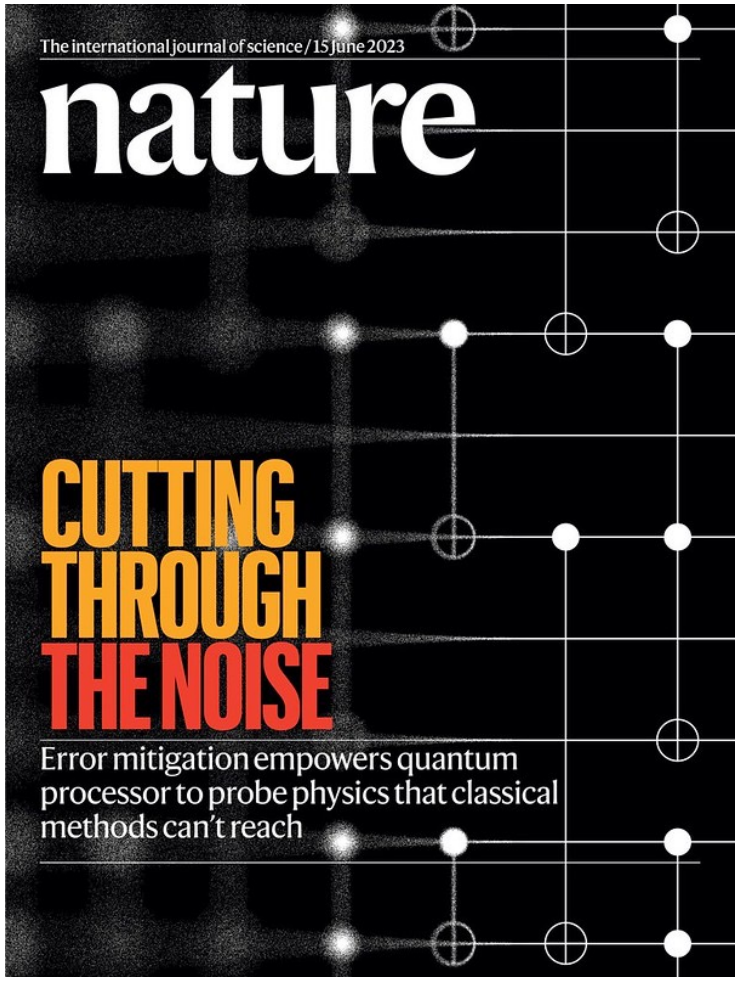
NISQ: Noisy Intermediate Scale Quantum

The era of quantum utility

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→ 2023

Evidence of utility before fault tolerance



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Quantum Utility → Quantum Advantage

Gates ⇨ 3K 5K 100M

PRE-UTILITY | ERROR MITIGATION | ERROR CORRECTION

Year ⇨ 2023 2024 2029

All circuits can be brute force run on classical hardware



Only approximate solutions exists and is the regime for exploring quantum advantage

Well in the error correction regime where today's known quantum algorithms exist




```
for _ in loop:
    qc,matrix,… = do_classical(param)
    A,… = SGESVD(matrix)
    job = sampler.run([qc])
    do_classical(job.result(), A, …)
    …
```

Thesis

The future of computing is mapping all problems to tensors and circuits

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Key milestones in bringing useful quantum computing to the world

2023

Establish quantum utility

5K operations on 100+ qubits



2026

Demonstrate quantum advantage

With clear separation and verifiability



2028

Demonstrate all key elements of a fault-tolerant quantum computer

Architecture, coupled logical modules, magic state factory



2029

Deliver the first fault-tolerant quantum computer

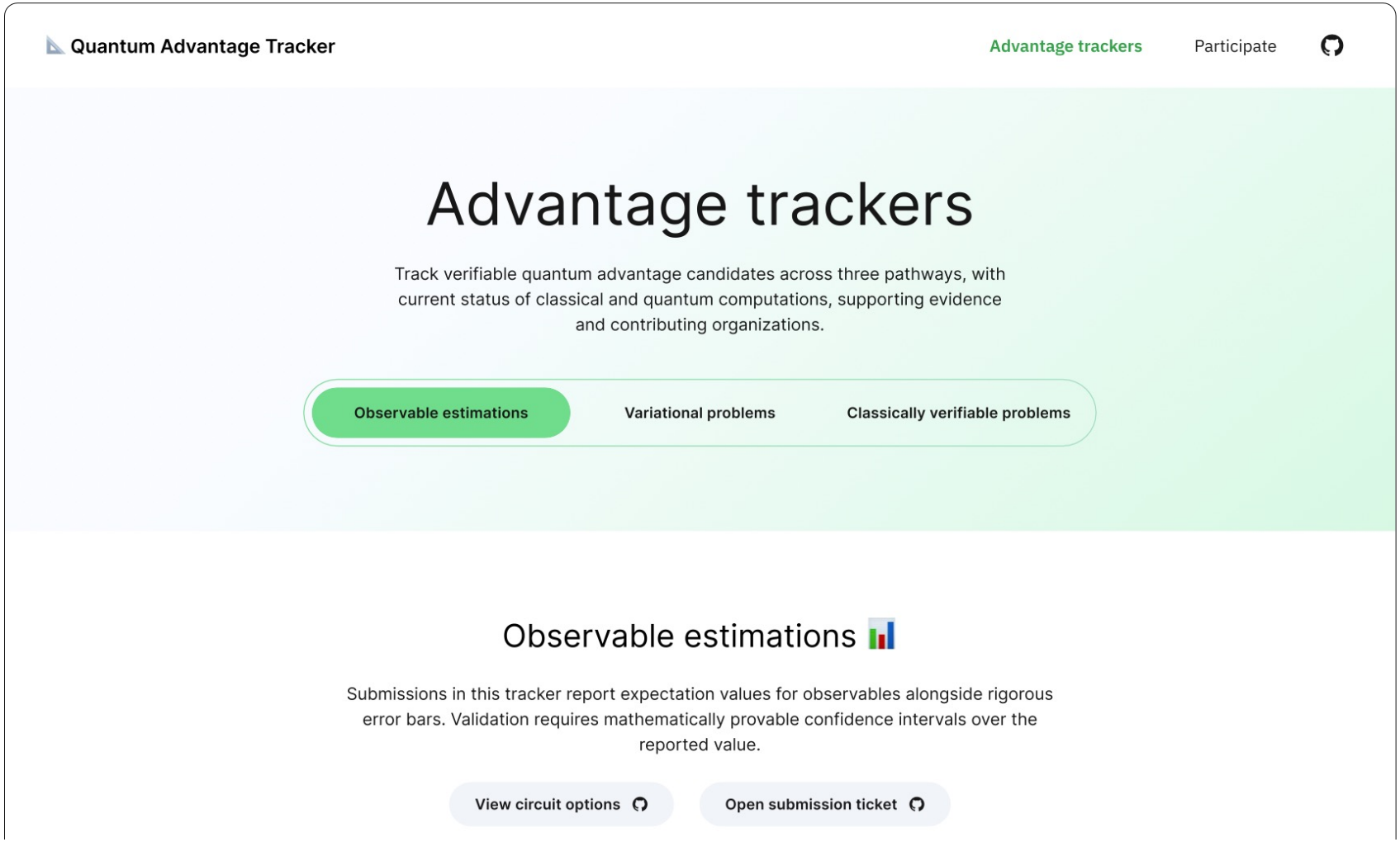
100M operations on 200 qubits



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2026 → Advantage

https://quantum-advantage-tracker.github.io



View circuit options Open submission ticket

Date ↑	Name	Method	Circuit	Exp. values...	Runtime	Compute resou...	Institution	Activity
2025-11-12	Submission name	Global Rescaling	Operator Loschmidth Echo_49_2400	0.7 [upper bound, lower bound]	Q: 2h C: –	Q: ibm_pittsburgh C: –	IBM Research, Algorithmiq, Flatiron	4
2025-11-12	Submission name	Belief propagation tensor network	Operator Loschmidth Echo_49_2400	0.98 [±1.1, -0.3]	Q: – C: 6h	Q: – C: 8x NVIDIA H100	IBM Research, Algorithmiq, Flatiron	5
2025-11-11	Submission name	Global Rescaling + partial PEC	Operator Loschmidth Echo_49_2400	0.68 [±1.1, -0.3]	Q: 10h C: –	Q: ibm_pittsburgh C: –	IBM Research, Algorithmiq, Flatiron	1
2025-11-10	Submission name	Global Rescaling + partial PEC	Operator Loschmidth Echo_49_2400	0.66 [±1.1, -0.3]	Q: 9h 6m C: –	Q: ibm_boston C: –	IBM Research, Algorithmiq, Flatiron	0
2025-11-09	Submission name	Pauli propagation	Operator Loschmidth Echo_49_2400	0.8 [±1.1, -0.3]	Q: – C: 24h	Q: – C: 32 core Intel...	Caltech	0

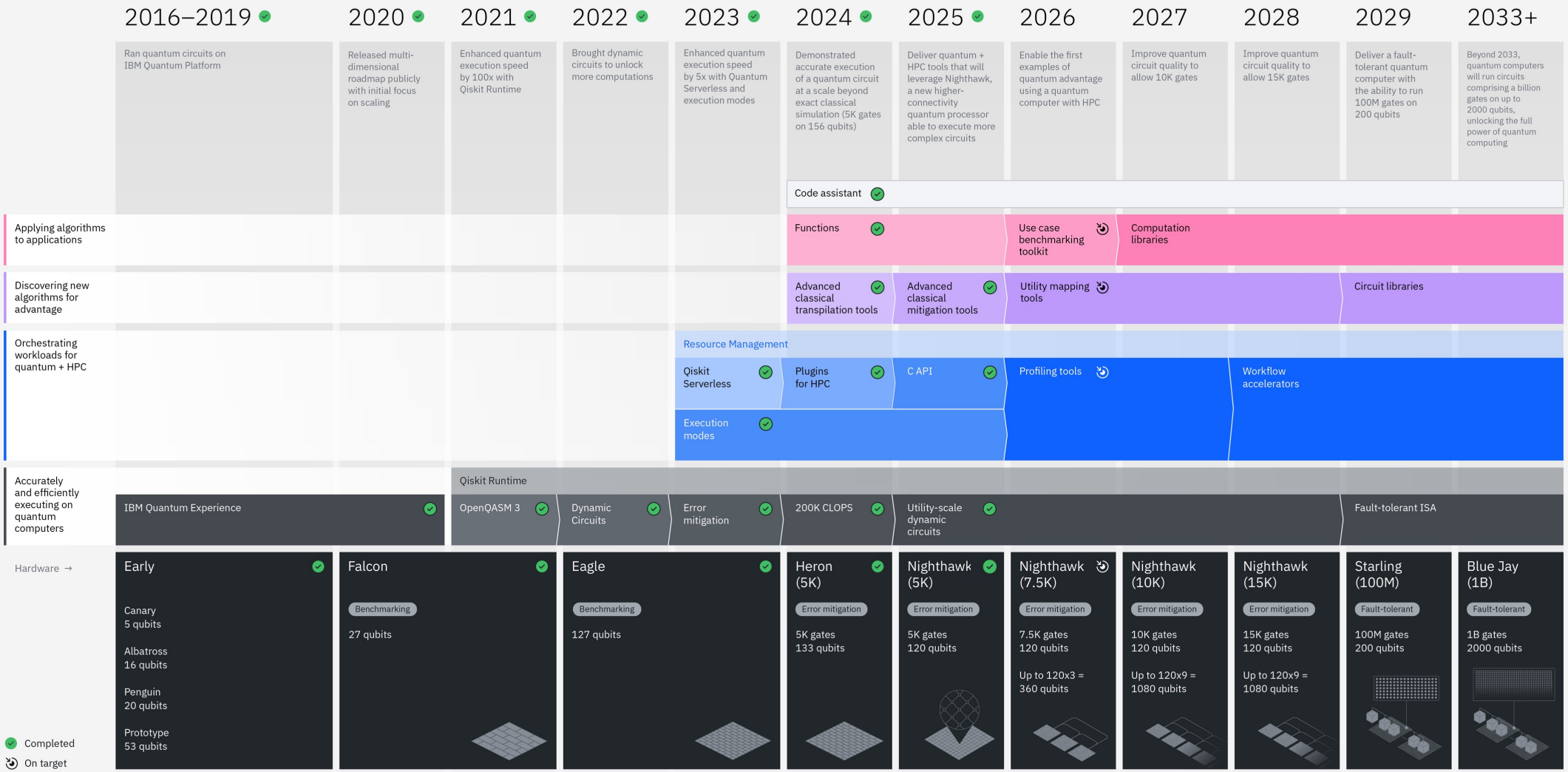
Participate

Have a quantum advantage candidate? Contribute it using the provided problem instances or your own circuit specifications, and help expand the list of verified candidates.

Learn more >

Development Roadmap

IBM **Quantum**



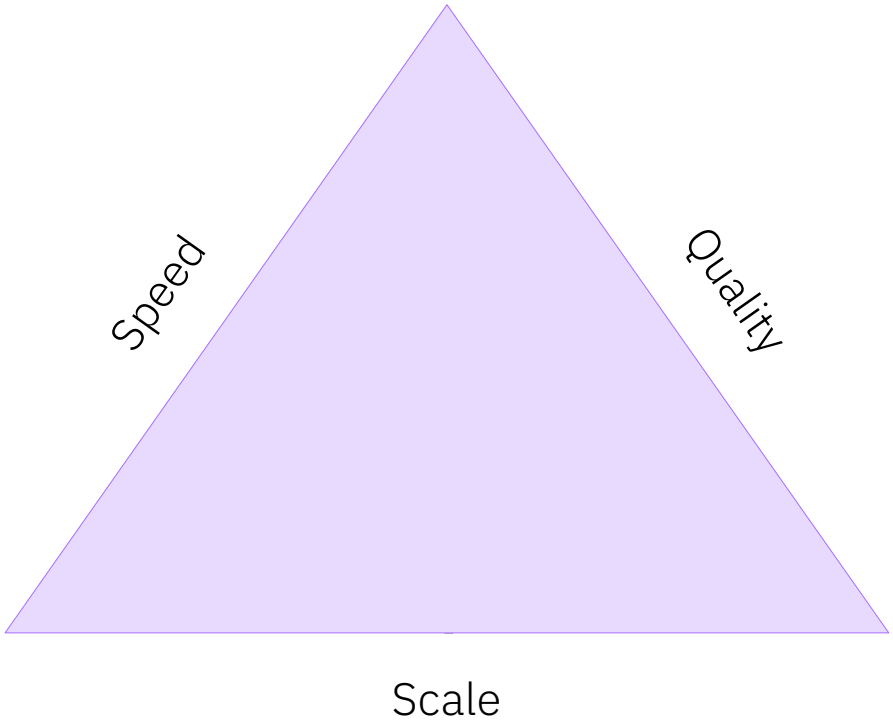
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Performance
= Scale + Quality + Speed

Scale
Stack enabled for utility scale computing

Quality
Consistent high quality across utility scale devices

Speed
Fast execution for cost effective results



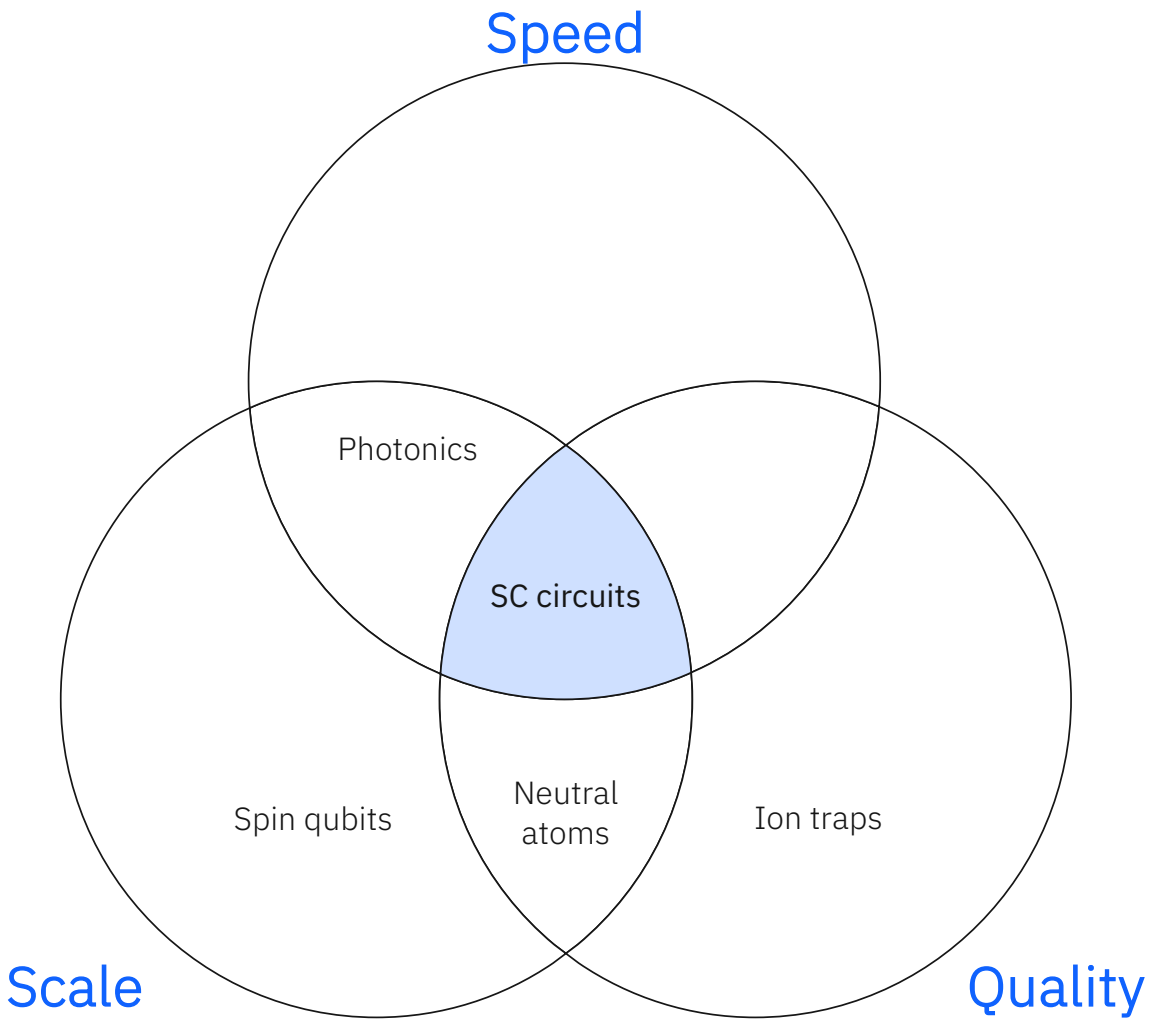
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Beware of hype

You need speed, scale, and quality — all three.

Some “full-stack” quantum computing companies are selling access to quantum hardware that may be too slow or inaccurate to do anything remotely useful.

IBM Quantum is the only full-stack quantum computing company that has demonstrated utility-scale hardware (>100 qubits), and we have made it available via the cloud.



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Introduction to Qiskit

1. What is Qiskit
2. Qiskit ecosystem

Qiskit

Release News: Qiskit v2.4 is here!

Technical release summary for Qiskit SDK v2.4, including updates on top new features, performance improvements, and our ongoing efforts to make Qiskit ready for fault-tolerant compilation.



1,300+ Qiskit certified developers

5,300+ Research papers using Qiskit

7,500+ Qiskit dependent projects

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Qiskit SDK

arXiv:2405.08810

The lingua franca of quantum computing; write once and execute quantum circuits on 10+ different hardware providers

- IBM Quantum
- AQT
- IQM
- Azure Quantum
- Alice & Bob
- IonQ
- Amazon Braket
- Superstaq
- Quantinuum
- Rigetti

Quantum SDK Preferred
(2024 Unitary Foundation Survey)

74%

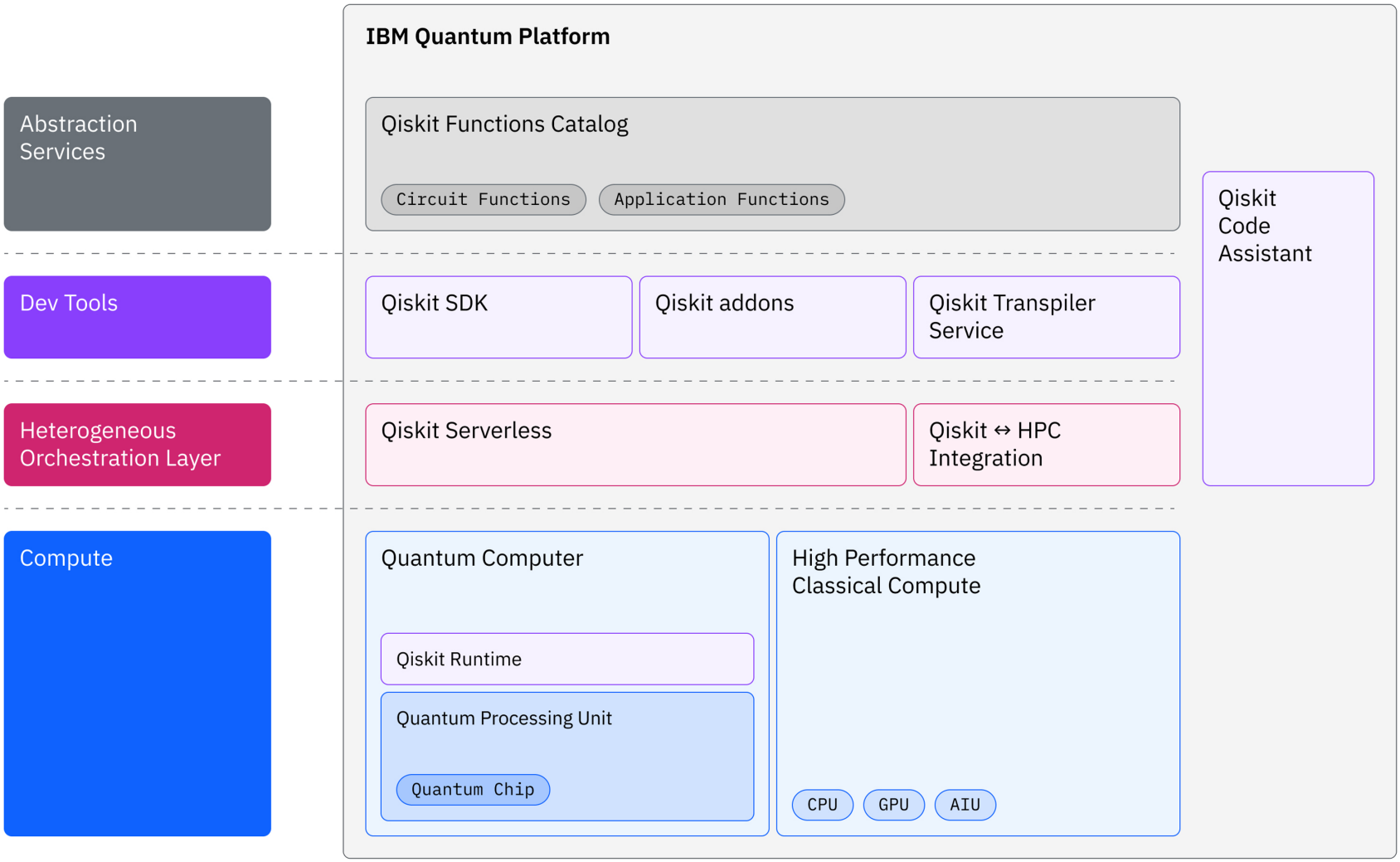
Qiskit contributors that are external to IBM

74%

Dependent Qiskit projects

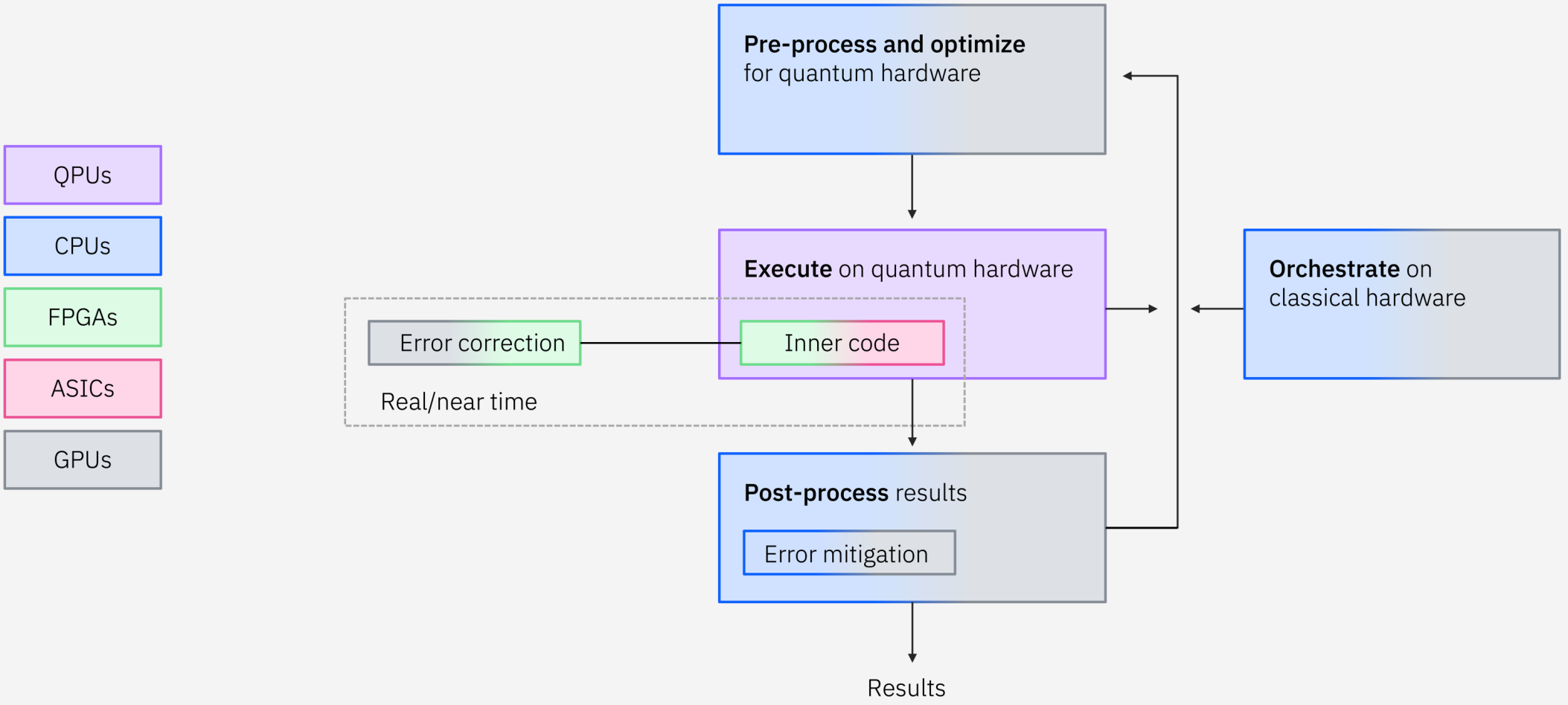
7400+

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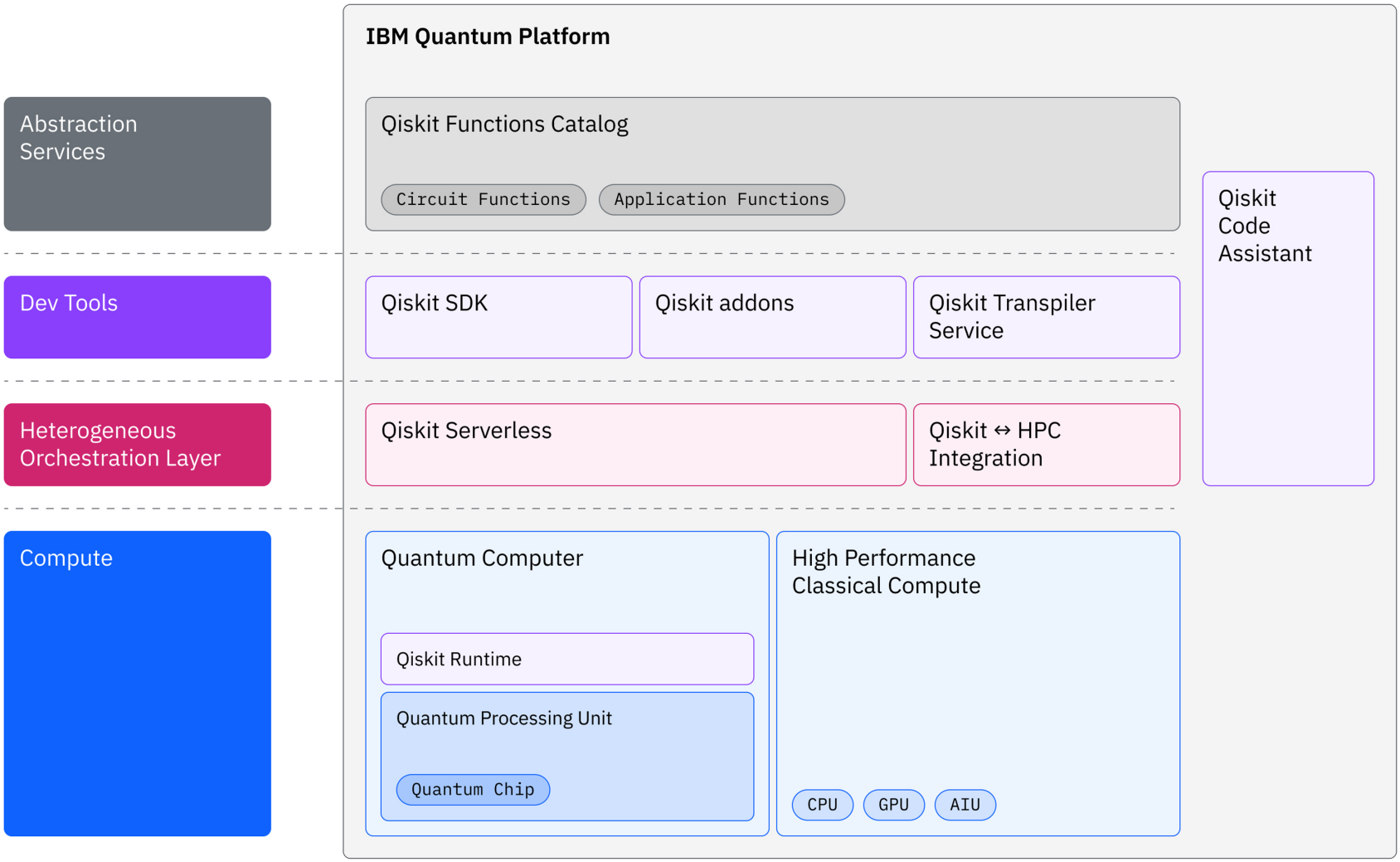


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Quantum-classical workflow



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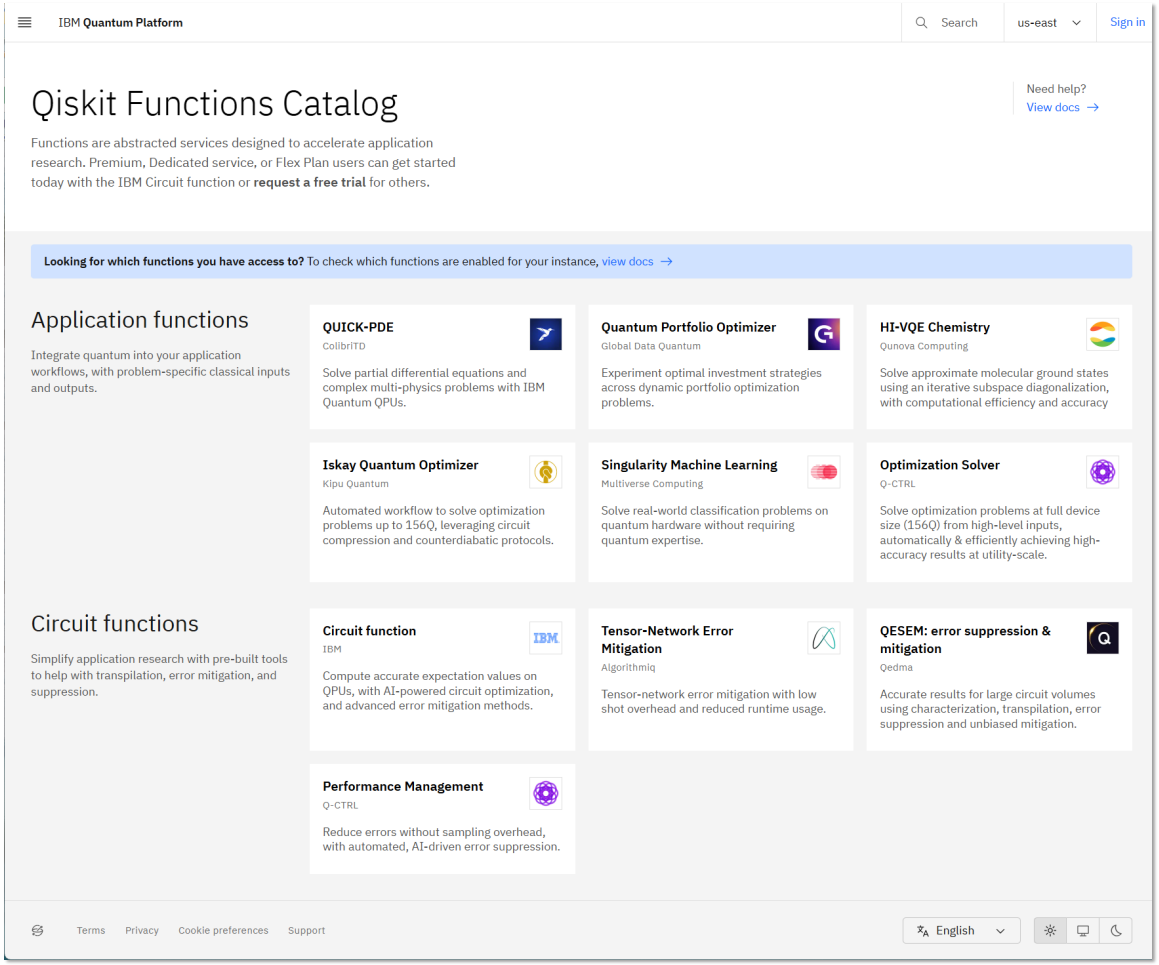
Qiskit Functions Catalog

During this first wave, Premium Plan users will be able to explore pre-built services from IBM and 3rd parties.

At launch, users will be able to leverage IBM Circuit Function out of the box.

Users will need to request access for 3rd-party functions, which could include free trials (at partner discretion).

https://quantum.cloud.ibm.com/functions



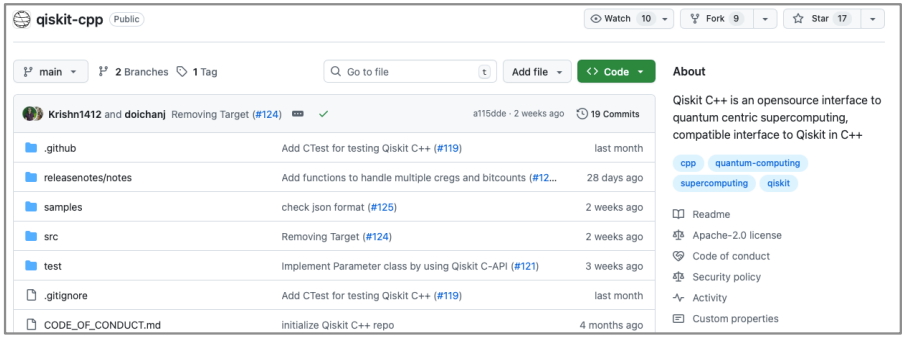
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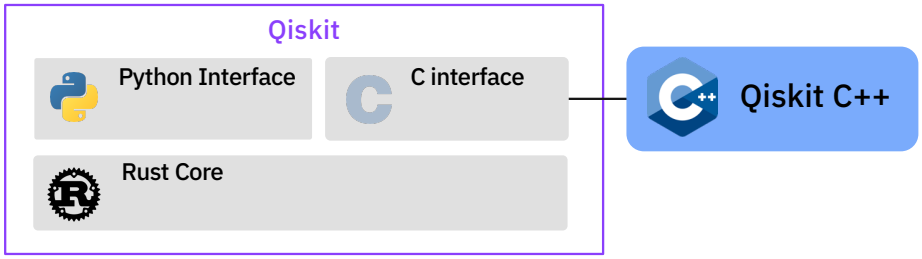
Introducing Qiskit C++

Laying the foundation of a C API

Qiskit 2.x introduced a C foreign function interface (FFI) to directly interface with Qiskit's internal Rust data model

C API can natively interface with almost every programming language (Fortran, C++, Rust, Python, Julia, etc.)





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AI enhancing quantum

AI-enhanced transpiler

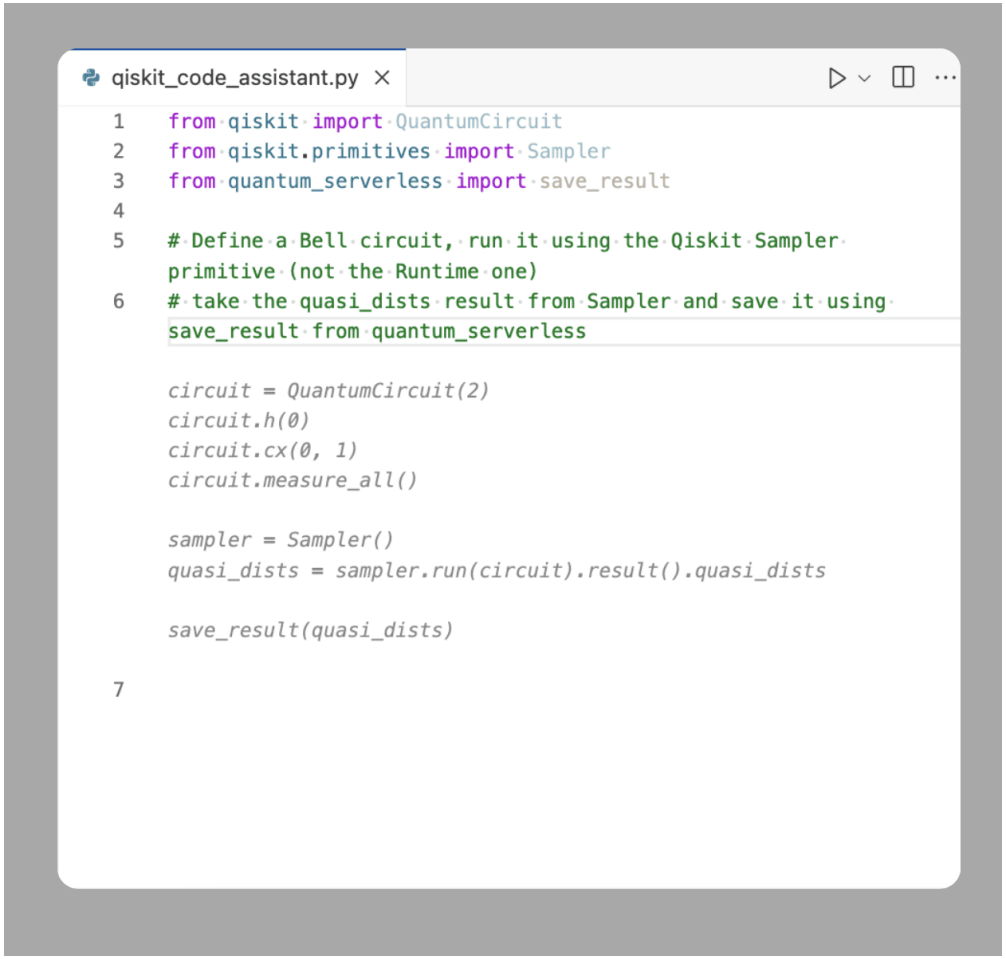
20–50% improvement in circuit depth and CNOT count compared to equivalent Qiskit transpiler passes

Qiskit Code Assistant

Collaboration with watsonx

IBM Granite foundation model based on and fine-tuned for Qiskit

Visual Studio Code extension



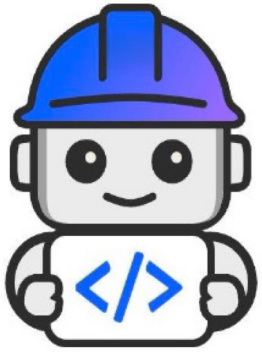
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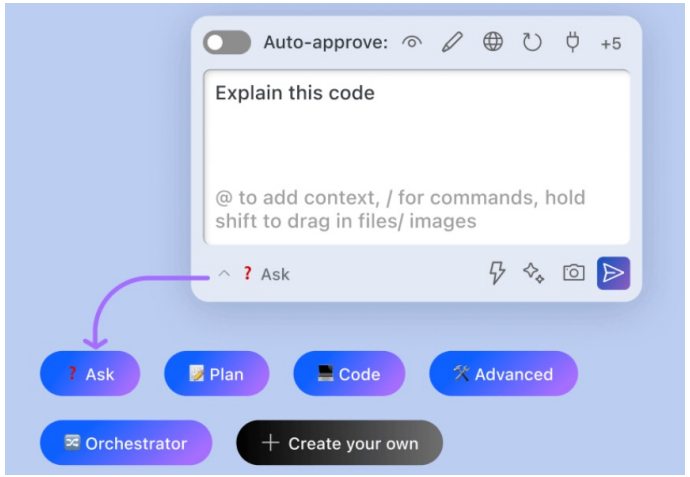
AI enhancing quantum | **IBM Bob**

Bob acts as a cognitive partner that understands your code and intent, assisting with **explaining Qiskit code**, **coding** it for you, **documenting** your code lines and more using natural language prompts.

IBM Bob supports the entire Software Development Lifecycle **from planning over coding to maintenance**. It supports a wide range of coding languages, incl. Python/Qiskit, Java, COBOL, Assembler, JavaScript, etc..

Key Capabilities of IBM Bob:
- AI-Powered Development (IDE)
- Modernization and Transformation
- Agentic Code Review & Security
- Documentation Generation
- Multi-Model Orchestration





If you want to try Bob for free, get the **30-day trial here**:
👉 ibm.biz/try_bob_for_free

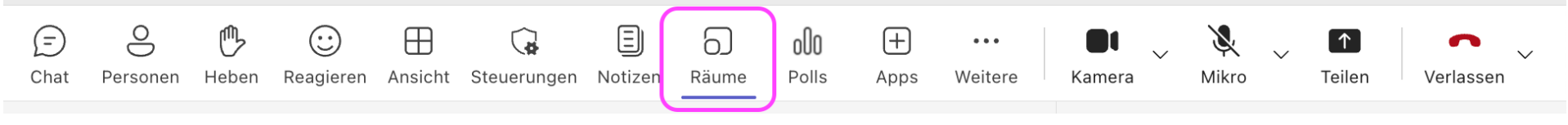
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Building Study Groups



It’s time to meet your study buddies…

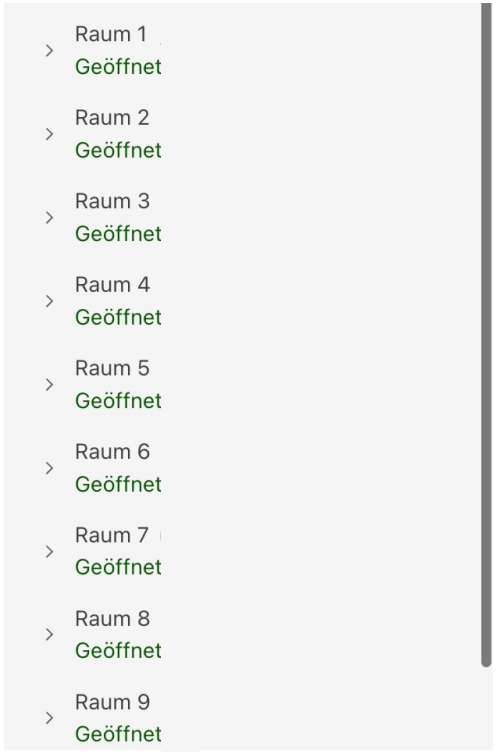
Building Study Groups





It's time to meet your study buddies…

Pick the "Raum"/Room with **your Study Group Number** that we sent you today



Q&A

Environment Setup

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
```

✓ 0.4s Python

```
from qiskit import QuantumCircuit

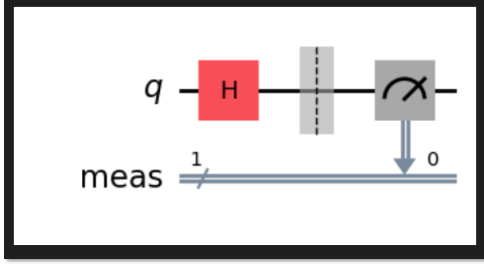
qc: QuantumCircuit = QuantumCircuit(1)

qc.h(qubit=0)

qc.measure_all()

qc.draw(output='mpl')
```

✓ 0.5s Python



Let‘s get your local environment ready…

Thank you!!

See you next Wednesday, same place, 13:00-14:30

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics